Project: Ant City Reproduction Encoder & Decoder
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Objective:

- Filter foreign ants.
- Store children ants order efficiently.
- Enable reconstruction of original queue.

Implementation:

- Encoder:
  - Reads queens and workers from txt.parents
  - For each worker, generates possible children by matching substrings with queen prefixes.
  - Checks input queue and removes foreign ants.
  - Saves filtered queue in binary format to minimize file size.
- Decoder:
  - Reads the binary file.
  - Prints children ants in the original order.

Design Decisions:

- Binary encoding of length + string reduces storage size.
- Checked substring matching to handle multiple children cases.
- Efficient memory usage: stores only valid children in vector before writing.

Results:

- Correctly filters foreign ants.
- Original queue can be reconstructed.
- File order size minimized compared to storing full strings in text.

Possible Improvements:

- Use dictionary compression for further reduction.
- Multi-threading for very large queues.

Conclusion:

- Both Encoder and Decoder work as required.
- File size optimization achieved with binary storage.